

Sarcoma QML Benchmark

Plain-language guide to the data files, the quantum methods, and the classical AI methods.
Prepared 2026-09-22 from the project spec and the raw data archive dated 2026-09-13.

Part 1: Every file and column, in simple words

One rule to remember: every file is a big table. In the clinical files, each row is a patient or a tumor sample. In the gene files, each row is a gene and each column is a sample.

What the ID codes mean

- **TCGA-3B-A9HI** is a patient.
- **TCGA-3B-A9HI-01** is that patient's tumor sample. The -01 means "primary tumor".

All files are in `.scratch/sarcoma-qml-benchmark/data/raw/2026-09-13_cbioportal/`.

The 5 main files

1. data_clinical_patient.txt (254 patients, 38 columns)

The patient information sheet. Only one or two columns are used for the task label. The rest are background.

Column(s)	What they mean
PATIENT_ID	The patient's unique code.
SUBTYPE	A rough sarcoma type label (LMS, DDLPS, etc.). It is blank for 26 patients, so we do not use it.
CANCER_TYPE_ACRONYM	Always "SARC" (sarcoma).
OTHER_PATIENT_ID	An old ID from a legacy system.
AGE, SEX, RACE, ETHNICITY, WEIGHT	Basic facts about the patient.
DAYS_TO_BIRTH	Days from birth to diagnosis. It is negative because the birth was before diagnosis.
AJCC_PATHOLOGIC_TUMOR_STAGE, AJCC_STAGING_EDITION, PATH_T_STAGE, PATH_N_STAGE, PATH_M_STAGE	Cancer staging: how big the tumor is (T), whether it reached lymph nodes (N), and whether it spread elsewhere (M).
ICD_O_3_HISTOLOGY	A code for the tumor's cell type. This is the label source. 8890/3 = leiomyosarcoma (LMS), 8858/3 = dedifferentiated liposarcoma (DDLPS).
ICD_10, ICD_O_3_SITE	Disease and body-location codes.

Column(s)	What they mean
HISTORY_NEOADJUVANT_TRTYN, RADIATION_THERAPY, PRIOR_DX	Treatment before surgery, radiation, and earlier cancers.
NEW_TUMOR_EVENT_AFTER_INITIAL_TREATMENT, PERSON_NEOPLASM_CANCER_STATUS	Whether the cancer came back, and whether the patient is tumor-free.
PRIMARY_LYMPH_NODE_PRESENTATION_ASSESSMENT	Whether the lymph nodes were checked.
DAYS_LAST_FOLLOWUP, DAYS_TO_INITIAL_PATHOLOGIC_DIAGNOSIS	Timing numbers.
OS_STATUS/MONTHS, DSS_STATUS/MONTHS, DFS_STATUS/MONTHS, PFS_STATUS/MONTHS	Survival results: overall, disease-specific, disease-free, and progression-free. Each has an alive/dead or event flag and a time in months.
FORM_COMPLETION_DATE, INFORMED_CONSENT_VERIFIED, IN_PANCANPATHWAYS_FREEZE	Paperwork and bookkeeping.
GENETIC_ANCESTRY_LABEL	Ancestry estimated from DNA.

2. data_clinical_sample.txt (255 samples, 19 columns)

This sheet describes the tumor sample itself.

Column(s)	Meaning
PATIENT_ID, SAMPLE_ID	The patient code, and the sample code that links to the gene files.
ONCOTREE_CODE	A short cancer-type code (LMS, DDLS, MFH, MFS, SYNS, MPNST, DES).
CANCER_TYPE, CANCER_TYPE_DETAILED	The cancer name. The detailed version says "Leiomyosarcoma" or "Dedifferentiated Liposarcoma". This is the second label source.
TUMOR_TYPE, SAMPLE_TYPE	Tumor and sample category (primary, recurrence, etc.).
GRADE	How aggressive the cells look. It is mostly blank here.
TISSUE_PROSPECTIVE / RETROSPECTIVE_COLLECTION_INDICATOR	Whether the tissue was collected before or after the study began.
TISSUE_SOURCE_SITE(CODE), TUMOR_TISSUE_SITE	Which hospital sent it, and where in the body it came from.
ANEUPLOIDY_SCORE	How abnormal the chromosome counts are.

Column(s)	Meaning
MSI_SCORE_MANTIS, MSI_SENSOR_SCORE	Two scores for "DNA copy-error repair is broken" (microsatellite instability).
SOMATIC_STATUS	Whether the mutation calling was done.
TMB_NONSYNONYMOUS	Tumor mutation burden: how many protein-changing mutations the tumor has.
TBL_SCORE	A count of broken chromosome pieces.

3. data_mrna_seq_v2_rsem.txt (gene activity, 20,531 genes x 253 samples)

- **Columns:** Hugo_Symbol is the gene name and is blank for some genes. Entrez_Gene_Id is a numeric gene ID. Then there is one column per tumor sample.
- **Values:** each number shows how active that gene is in that tumor. 0 means the gene is silent. Bigger numbers mean more active. This is called RSEM.
- **Preprocessing:** we take $\log_2(\text{value} + 1)$ so huge numbers do not drown out small ones.

4. data_cna.txt (copy-number, 25,128 genes x 253 samples)

- **Columns:** the same layout as file 3.
- **Values:** a gene can be copied too many or too few times in a tumor.

Value	Meaning
-2	Both copies gone.
-1	One copy lost.
0	Normal.
+1	Extra copy.
+2	Many copies (amplified).

- **In this data:** 0 is the most common, then -1 and +1. Both +2 and -2 are rare.
- **Why it matters:** DDLPS tumors nearly always have extra copies of a region on chromosome 12 (the MDM2 and CDK4 genes). That is a strong clue for telling DDLPS from LMS.

5. data_mutations_api.json (17,393 mutations from 234 samples)

A list of DNA typos, fetched through the cBioPortal API because the normal file was missing. Each entry has these fields:

Field	Meaning
sampleId, patientId	Whose tumor.
gene.hugoGeneSymbol, entrezGeneId	Which gene has the typo.
chr, startPosition, endPosition	Where on the DNA it sits.
referenceAllele -> variantAllele	The letter that was there, and what it changed to.

Field	Meaning
variantType	SNP, insertion, or deletion.
mutationType	The effect: Missense_Mutation (changes one amino acid), Frame_Shift_Del (breaks the protein), Silent, etc.
proteinChange	The protein-level change, for example M828*.
tumorAltCount, tumorRefCount	Reads showing the mutant and normal versions in the tumor.
normalAltCount, normalRefCount	The same counts in the patient's healthy blood. This is how we know the mutation is tumor-only.
center, mutationStatus, validationStatus, ncbiBuild, refseqMrnaId, keyword, studyId, molecularProfileId, uniqueSampleKey, uniquePatientKey	Lab, bookkeeping, and ID fields.

We will turn this into a yes/no table: one row per patient, one column per gene, and 1 if the gene has a protein-changing mutation.

The supporting files

gdc_crosscheck/gdc_manifest_{rnaseq,cna,maf}.tsv

Shopping lists from the GDC portal. They are not data. We used them only to check patient counts. Columns:

- **access:** whether the file is open or controlled.
- **cases.0.case_id, cases.0.submitter_id:** the patient IDs.
- **data_category, data_format, experimental_strategy:** what kind of file it is.
- **file_id, file_name, id:** the file identity.

publication_tables/

- **SARC_Master_Patient_Table.xlsx** covers the 206 patients from the 2017 paper, with 55 named columns. It is not our cohort. Its columns are:

Tumor and clinical	Histology, age, gender, tumor size and site, tumor grade (FNCLCC_grade), mitotic rate, necrosis, and recurrence and survival fields.
Chromosome-12 flags	AMP_12q15_MDM2 / CDK4 / HMGA2 / DDIT3 / FRS2. These matter for DDLPS.
Cluster labels	One column per data type (CN_cluster, methylation_cluster, mRNA_cluster, and so on).
Tumor-quality numbers	purity (share of cancer cells), ploidy (DNA amount), leukocyte_fraction (immune cells), Mutation_Load.

- **firehose_clinical.tsi** is the 2016 snapshot with 261 patients. It is where 105 LMS came from.

- **lms_ddlps_patient_ids.txt** holds the 164 patient IDs, with a class label (LMS or DDLPS) after each ID.
- **CHECKSUMS.sha256** is a fingerprint per file, used to detect changes. **NOTES.md** is the logbook.

What we actually use for the ML: the label comes from ICD_O_3_HISTOLOGY. The features come from files 3, 4 and 5. Everything else, especially survival and staging, is background and stays out of the model.

Part 2: Quantum and classical AI parts

The core problem

A tumor has about 20,000 gene measurements. We have about 160 patients, which is very few. A quantum circuit can only take about 8 to 16 numbers, one per qubit. So both sides start with the same three steps, done fresh inside every cross-validation fold so that test patients never influence them:

1. MAD filter	Keeps the 1,000 genes that vary most across patients. Genes that barely change carry no information.
2. ElasticNet	A simple linear model with a penalty that pushes useless genes to exactly zero. What survives is the final 8 or 16 genes.
3. Scaling	Rescales the numbers. Classical models get standard or 0-to-1 scaling. Quantum models get 0 to π .

The quantum side

Encoding ("mapping") means loading numbers into qubits

The spec's method is **angle encoding**: each gene value becomes the rotation angle of one qubit. That is why we scale to 0 to π . A rotation of 0 leaves the qubit at "0". A rotation of π flips it fully to "1". Going past π would start to repeat earlier angles, so $[0, \pi]$ uses the full range without overlap.

Honest note: the spec fixes "angle encoding to $[0, \pi]$ ", but it does not say which rotation gates to use or how the qubits get entangled inside the mapping. You called it "custom" mapping, and that design choice is still open.

Technique 1: fidelity quantum kernel (FQK), a stretch model

- A kernel is a similarity score between two patients.
- Here the score is how much the two patients' qubit states overlap.
- With more qubits, every pair of patients ends up looking equally different. The kernel becomes useless. This is called exponential concentration.

Technique 2: projected quantum kernel (PQK-SVM), a core model

- Encode the patient, then measure each qubit separately. This gives ordinary numbers.
- Compute a normal similarity score on those numbers.
- Feed the scores to an SVM, a classifier that draws the best boundary between two groups.
- This avoids the concentration problem.

Technique 3: variational circuit or hybrid QNN (VQC), a core model

- The circuit is: encode the data, then apply trainable rotation layers (the "EfficientSU2" pattern), then measure.
- The spec says 8 qubits, 2 entangling layers, and 32 trainable numbers.
- A normal computer adjusts those 32 numbers with Adam, a common optimizer, to reduce mistakes.

- **Parameter-shift rule:** to find which way to nudge a number, run the circuit twice with that number shifted slightly up and slightly down, and compare the results. This works on real quantum hardware, where backpropagation is not possible.
- **Check when we build it:** I believe the standard Qiskit setup gives 48 parameters at 2 layers and 8 qubits, not 32. It only hits 32 if the final rotation layer is skipped. I have not verified this against the installed library.

Technique 4: noise simulation, a stretch goal

This adds realistic hardware errors to a simulator. It is **not** a real quantum computer run, and every writeup must say that.

The classical AI side

Model	Simple explanation
MLP (core)	A small neural network: layers of numbers that learn patterns. It is the "deep learning" opponent. With about 130 training patients it tends to memorize instead of learn. That is exactly what the study wants to test.
RBF-SVM	The classical twin of the quantum kernels. It uses the same SVM with an ordinary similarity score. If the quantum kernel does not beat it, the quantum part is not adding anything.
XGBoost	Builds many small decision trees, each fixing the last one's mistakes. Stretch model.
TabNet	A neural net that learns which features to look at. Stretch model.

How results are judged

Why not plain accuracy: always guessing "LMS" is right about 64% of the time (105 of 163). So we use:

Metric	What it tells you
PR-AUC (primary)	How well the model finds the rarer class without false alarms.
ROC-AUC	Overall ranking quality.
Balanced accuracy	Accuracy averaged per class.
MCC	One fair summary number that punishes lopsided guessing.

- **Cross-validation:** split the patients into 5 groups, train on 4, test on the 5th, rotate, and repeat this with different random splits.
- **Corrected t-test:** the repeated splits share patients, so an ordinary test would be overconfident. The corrected version accounts for that.
- **One planned test:** best quantum model versus the MLP on PR-AUC. A null result ("no difference") counts as equally valid.

- **Experiment 2:** shrink the training set to 20, 40, 80, and 130 patients to see when the neural net breaks down.

Spec check: it says "5x5 (50-fold)" cross-validation, but 5 repeats x 5 folds is 25, not 50. We should decide which one is meant.

One correction to something I said earlier

I said "all sources are open access, no credentials needed." That is only proven for the cBioPortal files we downloaded. I only downloaded GDC file lists, never the GDC files themselves. Those lists show most GDC files are controlled-access:

Data type	Open files	Total files
Mutation files	238	1,512
RNA-seq	265	2,385
Copy-number	2,304	3,364

The open ones may be enough for the 5 missing LMS patients if you choose the N=163 option. I have not checked that. It should be verified before that choice is made.